

Software Requirement Specification (SRS)

English to Hindi Language Translator Web Application

1. Introduction

Purpose: This project develops a web-based application that translates English text into Hindi using Natural Language Processing (NLP).

The system uses the Helsinki-NLP/opus-mt-en-hi translation model to perform machine translation.

Users can enter English text and receive Hindi translation instantly through a web interface.

1.2 Scope

The system allows users to translate English sentences into Hindi using a web application.

Functions include accepting English text input, processing translation using an NLP model, and displaying Hindi output.

The application is built using Python Flask and deployed online using Render.

1.3 Definitions and Acronyms

NLP – Natural Language Processing

SRS – Software Requirement Specification

UI – User Interface

API – Application Programming Interface

MT – Machine Translation

2. Overall Description

The system is a standalone web application that performs translation using a pre-trained NLP model.

Architecture:

User Browser → Frontend (HTML/CSS/JS) → Flask Backend → Translation Model → Hindi Output

2.2 Product Functions

- Accept English text input
- Translate English to Hindi

- Display translated Hindi text
- Handle invalid or empty input

2.3 User Classes

General User: Any person who wants to translate English text into Hindi.

No authentication is required in the current version.

2.4 Operating Environment

Backend: Python

Framework: Flask

NLP Library: Transformers

Model: Helsinki-NLP/opus-mt-en-hi

Frontend: HTML, CSS

Hosting: Render

3. System Features

Text Input Feature: Users enter English text in the input box.

Translation Feature: The system processes input using the NLP translation model.

Example:

Input: Hello, how are you?

Output: ■■■■■■■■, ■■ ■■■■■ ■■■■?

Result Display: The translated Hindi text appears on the webpage.

4. External Interface Requirements

User Interface includes:

- Title: English to Hindi Translator
- Text input box
- Translate button
- Output section

Users only need a computer or smartphone with internet access.

5. Non-Functional Requirements

Performance: Translation time should be less than 3 seconds.

Security: No personal data stored; input validation required.

Usability: Simple interface usable on desktop and mobile.

Reliability: System should handle unexpected input and run continuously when deployed.

6. System Architecture

Architecture Type: Client-Server

Client (Browser) → Flask Server → Translation Model

7. Future Enhancements

- Hindi to English translation
- Multiple language support
- Voice input
- Text-to-speech output
- User login system
- Translation history

8. Conclusion

The English to Hindi Language Translator WebApp provides an efficient method to translate English text into Hindi using NLP.

The application uses the Helsinki translation model, runs on a Flask backend, and is deployed using Render to make it accessible online.